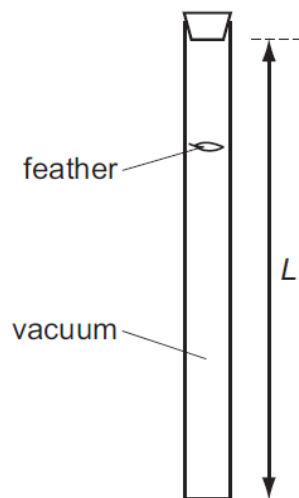


- 4 The diagram shows a laboratory experiment in which a feather falls from rest in a long evacuated vertical tube of length L .



The feather takes time T to fall from the top to the bottom of the tube.

How long will the feather take to fall $0.50 L$ from the top of the tube?

- A** $0.25 T$ **B** $0.29 T$ **C** $0.71 T$ **D** $0.75 T$

Ans: C

$$s = ut + 0.5at^2$$

For 1st half,

$$0.5L = 0.5at_1^2 \quad \text{--- (1)}$$

For entire journey,

$$L = 0.5aT^2 \quad \text{--- (2)}$$

Solve (1) and (2)

$$t_1 = 0.71T$$

- 5 The diagram shows two blocks of mass m and $2m$ connected by a light inextensible cord passing